

Introduction to Human Physiology

For Human Nutrition students

Anatomy and Physiology

Physiology vs Anatomy

Explain: *Function vs Structure*

Levels of Organization

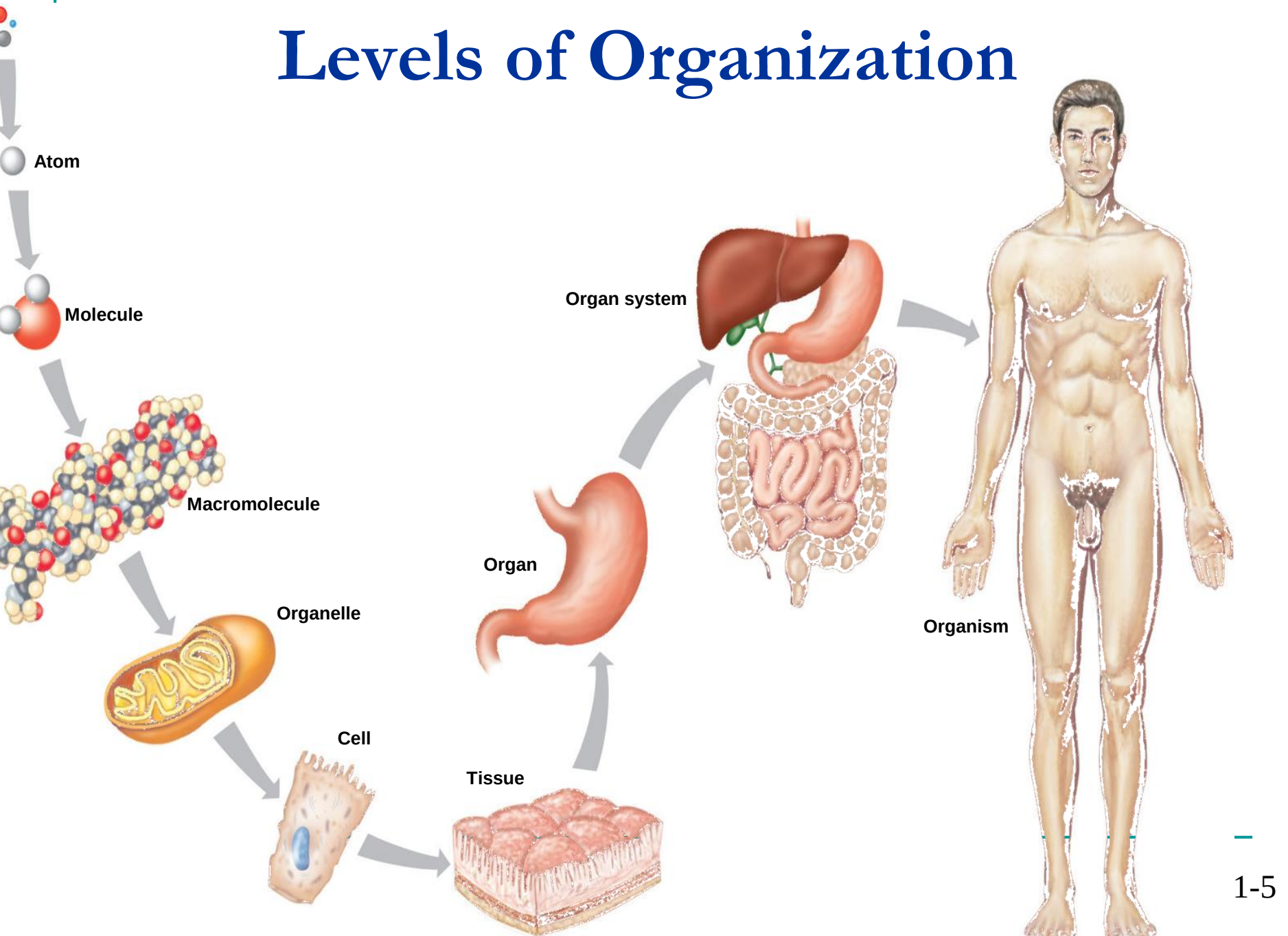


TABLE 1.2

Organ System	Major Functions
Integumentary	Protect tissues, regulate body temperature, support sensory receptors
Skeletal	Provide framework, protect soft tissues, provide attachments for muscles, produce blood cells, store inorganic salts
Muscular	Cause movements, maintain posture, produce body heat
Nervous	Detect changes, receive and interpret sensory information, stimulate muscles and glands
Endocrine	Control metabolic activities of body structures
Cardiovascular	Move blood through blood vessels and transport substances throughout body
Lymphatic	Return tissue fluid to the blood, carry certain absorbed food molecules, defend the body against infection
Digestive	Receive, break down, and absorb food; eliminate unabsorbed material
Respiratory	Intake and output of air, exchange of gases between air and blood
Urinary	Remove wastes from blood, maintain water and electrolyte balance, store and transport urine
Reproductive	Produce and maintain sperm cells, transfer sperm cells into female reproductive tract Produce and maintain egg cells, receive sperm cells, support development of an embryo and function in birth process

Characteristics of Life

- Movement
- Responsiveness
- Growth
- Reproduction
- Respiration
- Digestion
- Absorption
- Assimilation
- Circulation
- Excretion

Organ Systems

Integumentary

Skeletal

Muscular

Nervous

Endocrine

Cardiovascular

Lymphatic

Digestive

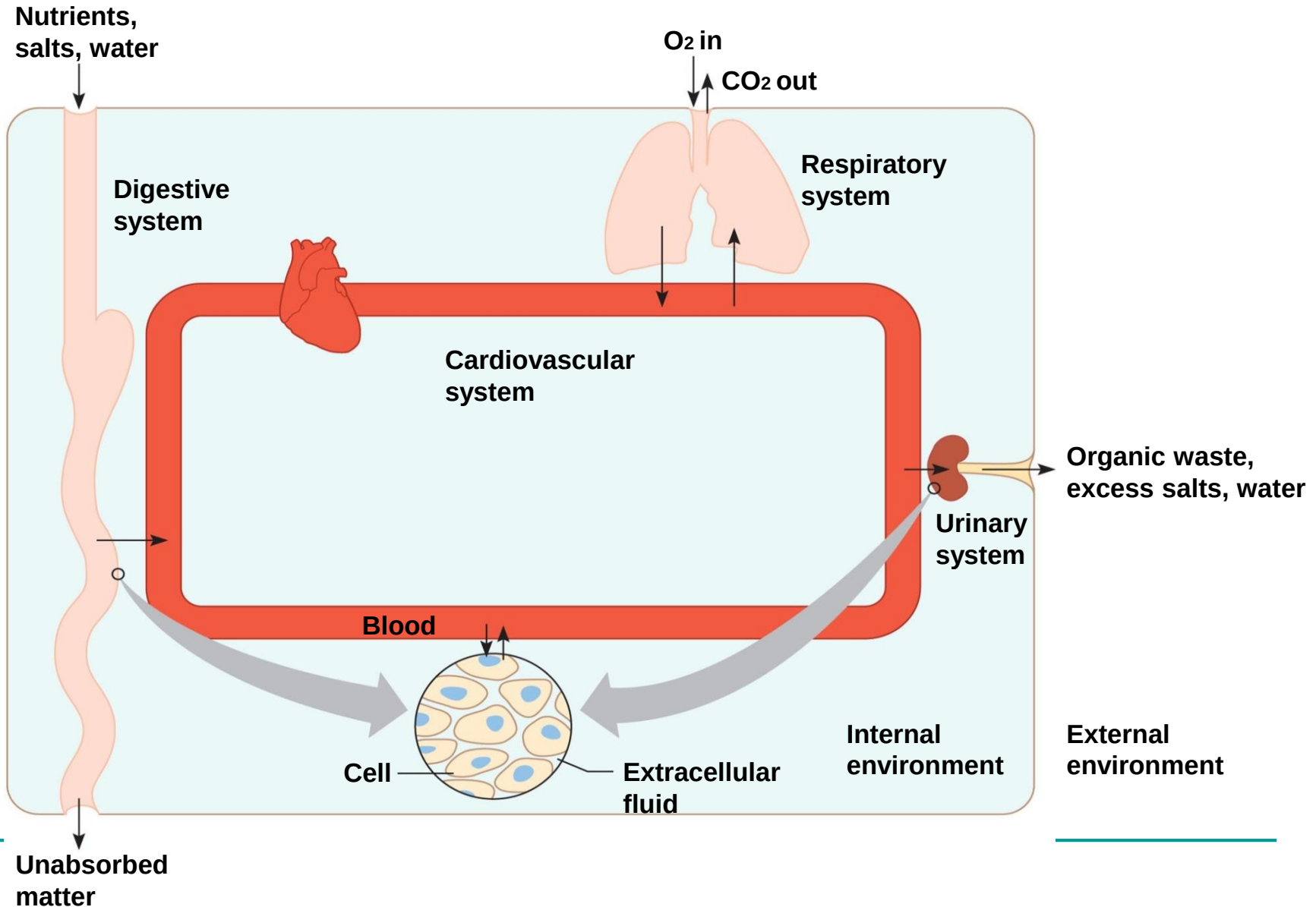
Respiratory

Urinary

Reproductive

Which organ system(s) contribute to each of the characteristics listed on the left?

Organ Systems Are Inter-related



Requirements of Organisms

Water: most abundant substance in body

- metabolism
- transport
- regulates body temperature

Food

- energy
- building blocks

Oxygen - ~20% of air

- releases energy from nutrients

Heat

- helps control rate of metabolic reactions

Pressure

- atmospheric pressure – breathing
- hydrostatic pressure – blood flow

Homeostasis

Imagine one day is very cold, while the next day is very hot.

What effect does that have on your body temperature?

External environment – keeps changing

Internal environment - remains stable.

How?

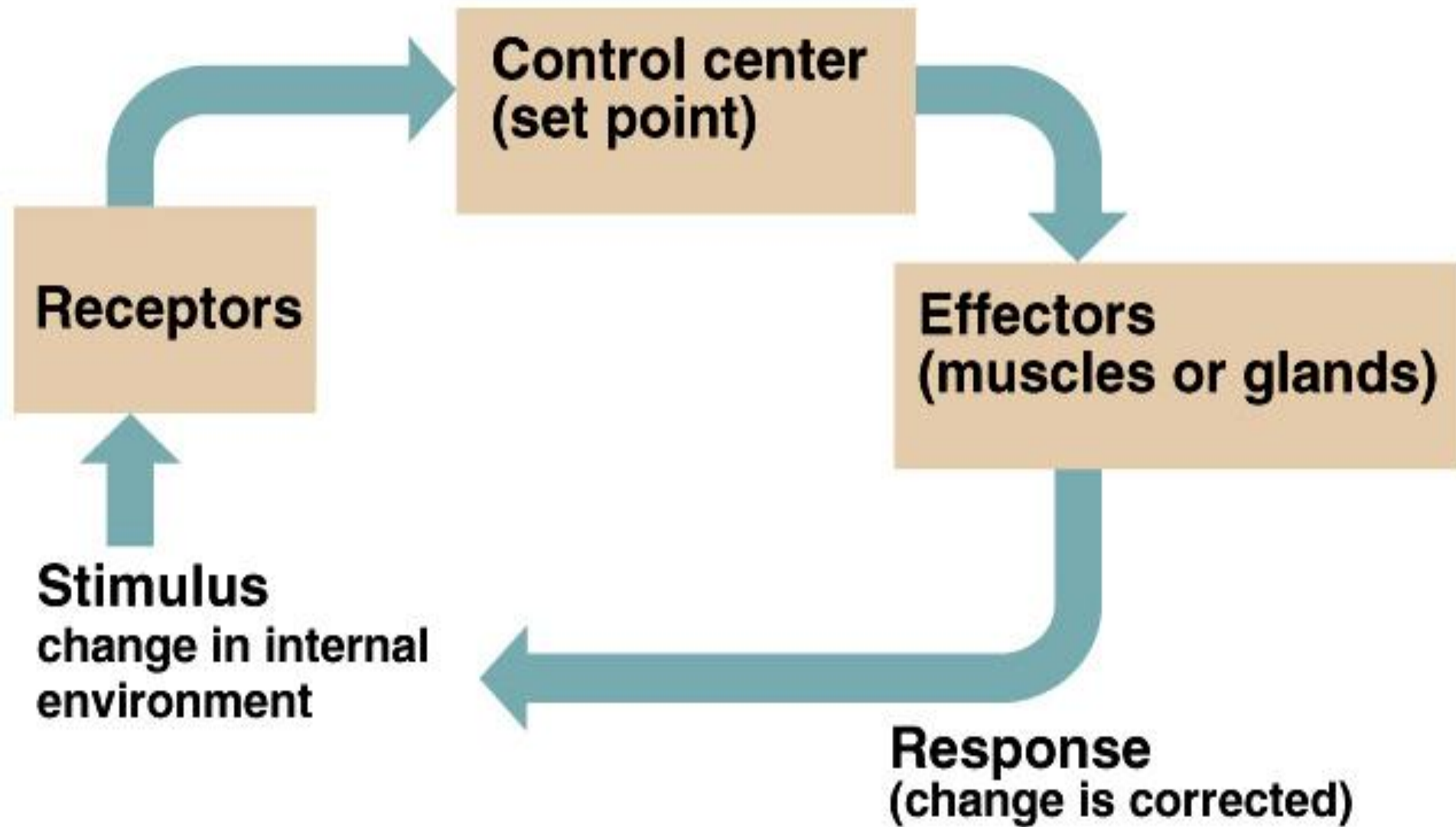
Body has “Homeostatic Mechanisms”

Homeostatic Mechanisms

- monitor internal environment
- corrects changes

3 Components of Homeostatic Mechanisms:

- *receptors*
 - *control center*
 - *effectors*
-



Types of Regulation

- **negative feedback**

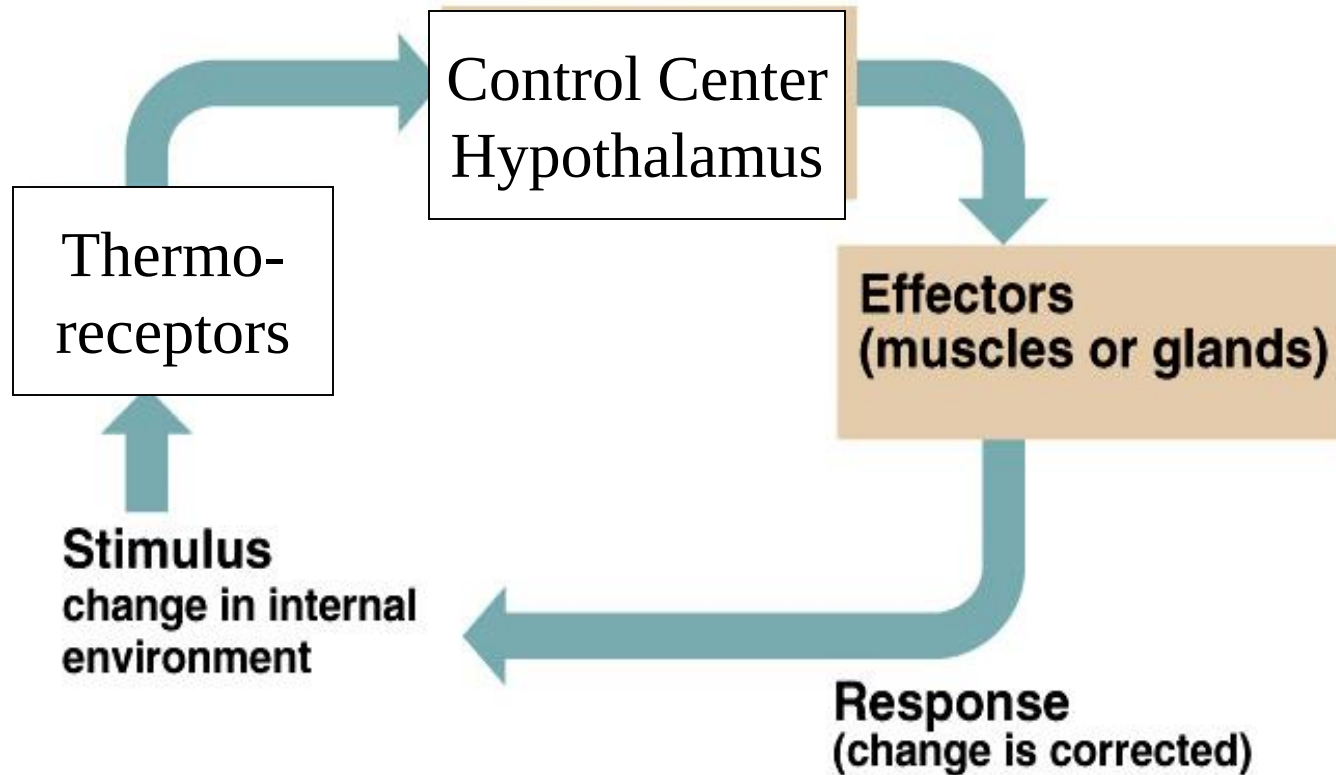
- **most common!**

- Example. maintaining body temperature; release of most hormones, etc.

- **positive feedback**

- blood clotting

Example of Homeostasis – Control of Body Temperature



What temperature does your body try to maintain?

What changes occur when your body temperature is:
- too high? - too low?

Integumentary System

2nd

What is the Integument?

- The INTEGUMENTARY SYSTEM includes all the structures and tissues related to the skin.
- This includes:
 1. skin
 - a. Epidermis
 - b. Dermis
 2. accessory organs
 - a. Hair
 - b. Nails
 - c. Glands

...

- The skin covers the entire body and provides protection against injuries, infection, and toxic compounds.
- The system contains many nerve endings that act as RECEPTORS for pain, temperature, touch, and pressure.

Anatomy of the Skin

- Upper layer of skin is called EPIDERMIS
EPI- means on top of
DERMIS means skin
- The lower layer of skin is called the DERMIS.

Epidermis

- Is multilayered.
- Contains is superficial layer of dead skin cells called STRATUM CORNEUM.
- The stratum corneum is composed of cells called KERATIN.
- Is AVASCULAR (no blood vessels)
- Has no nerve tissue (can't feel pain)

Dermis

- Has two distinct layers:
 1. Papillary Layer (more superficial)
 2. Reticular Layer (deeper)
- Accessory organs begin the dermis.
- Some of the accessory organs end or cross into the epidermis.
- The dermis contains blood vessels and nerve tissue.

Subcutaneous Fat and Superficial Fascia

- The SubQ fat is found deep to the dermis.
- It contains the larger blood vessels of the skin.
- It is composed of adipose (fat) tissue, a form of connective tissue.
- Deep to the subQ layer is the fascial layer that covers the muscles beneath.

....

Remember, the dermis ...
contains the hair
follicles, sweat
glands, and
sebaceous glands (oil
glands)

Dermat/o/scler/osis



Therefore, **dermatosclerosis** is literally translated as a disease that results in the hardening of the skin. Also known as **scleroderma**, this disease is an autoimmune disease that results in scarring of the vital organs resulting in death.

...

- The oil secreting glands of the skin are called the SEBACEOUS GLANDS.
- The sweat glands are known as SUDORIFEROUS GLANDS.

Mycodermatitis

fungus

skin

inflammation



A mycodermatitis is an inflammatory disease of the skin caused by a fungus. A common example of this is ATHLETE'S FOOT or TINEA PEDIS.

...

The pigment that causes skin coloration is
MELANIN

Melanosis is the abnormal condition of
black.

A melanoma is a tumor of black pigment.

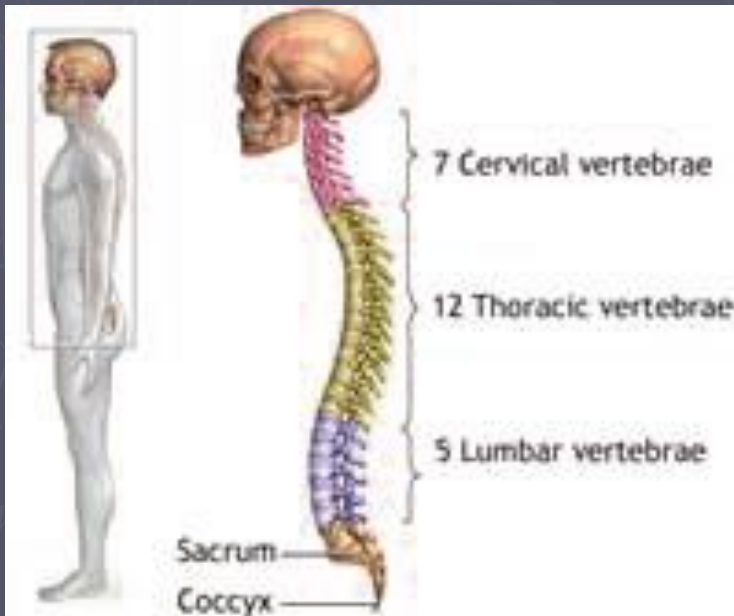
Skeletal System

3rd



What are the 5 Functions of the Skeletal System?

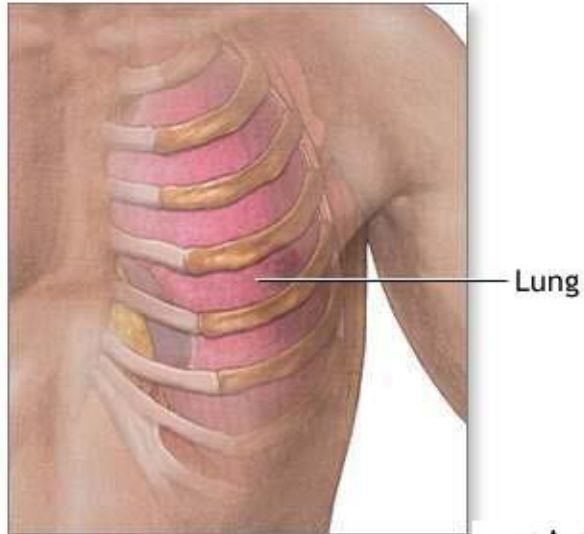
Muscle attached to bones!!



1. Movement: Skeletal system provides points of attachment for muscles. Your legs and arms move when the muscles pull on the bones.

2. Support: The backbone is the main support center for the upper body. It holds your head up and protects your spinal cord.

What are the 5 Functions of the Skeletal System?



ADAM.

3. **Protection:** The bones of your skull protect your brain. Your ribs protect your lungs and heart from injury.

4. **Makes Blood:** Red and white blood cells are formed by tissue called marrow, which is in the center of the bone.



What are the 5 Functions of the Skeletal System?

Calcium helps build
dense bones



ADAM.

5. Storage: Bones store minerals, such as calcium and phosphorus, for use by the body

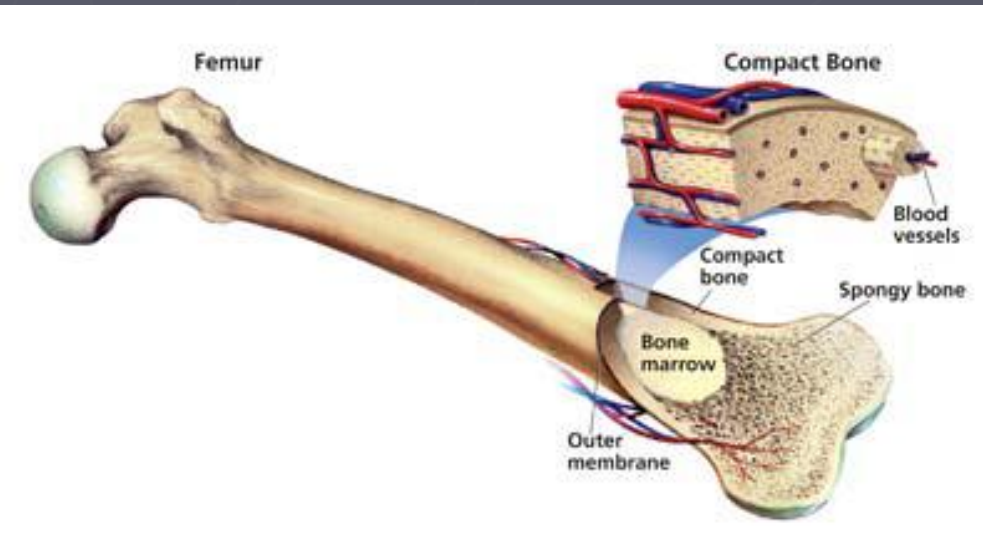
What are the Two Major Skeletal System Parts?



- **Axial Skeleton:** The axial skeleton includes the skull, spine, ribs and sternum.
- **Appendicular Skeleton:** The appendicular skeleton includes the appendages of the body, which are the shoulders, arms, hips, and legs.

What is the Structure of Bone?

- In the spaces of many bones there is a soft connective tissue called **marrow**.
- **Two types of Marrow:**
 - ▶ Red
 - Red Marrow produces most of the body's blood cells.
 - ▶ Yellow
 - Yellow Marrow stores fat that can serve as an energy reserve.



How Strong are Bones?



- The structure of bones make it both strong and light weight.
- 20 % of an adults body weight is bone.
- Bone is made up of 2 minerals:
 - ▶ Calcium
 - ▶ Phosphorus



Thanks .



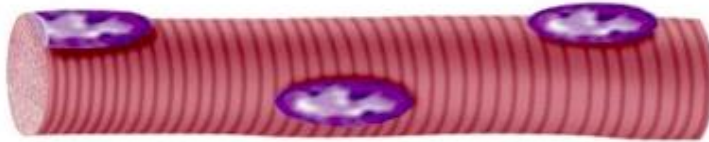
Muscular System

4th

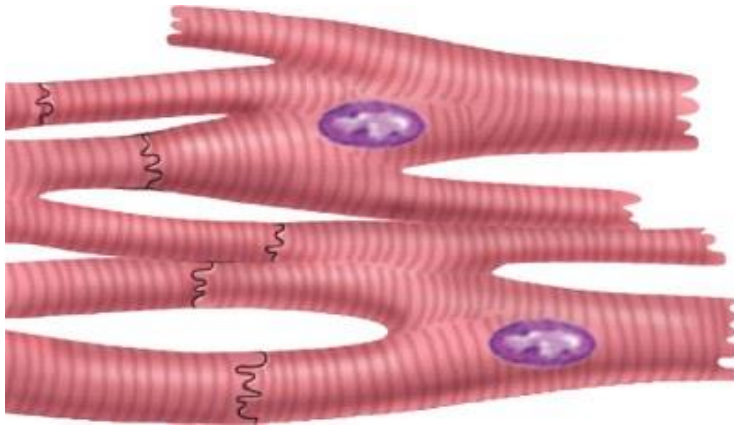
Functions of Muscular Tissue

- Like nervous tissue, muscles are excitable or "irritable"
 - they have the ability to respond to a stimulus
- Unlike nerves, however, muscles are also:
 - **Contractible** (they can shorten in length)
 - **Extensible** (they can extend or stretch)
 - **Elastic** (they can return to their original shape)

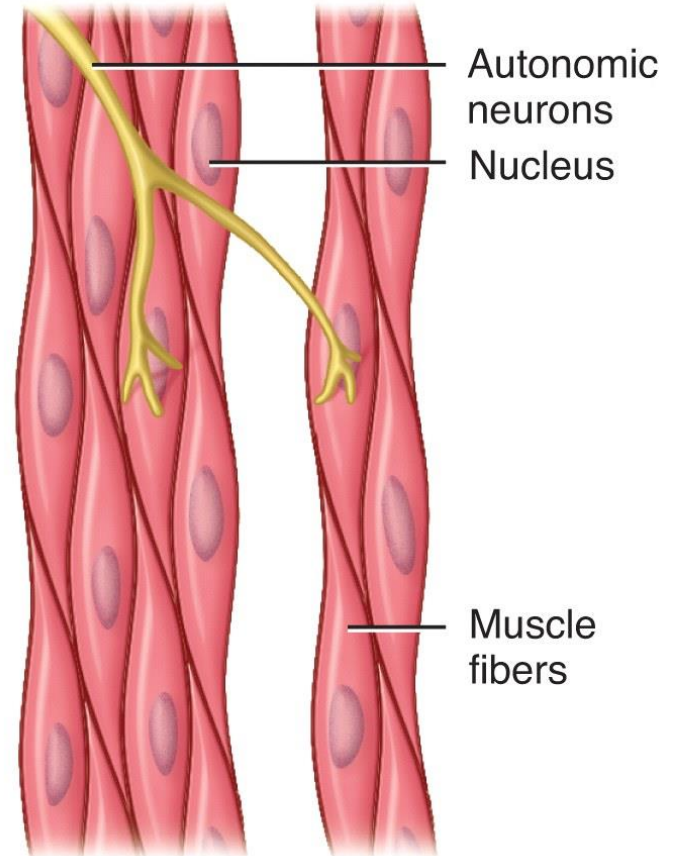
Three Types of Muscular Tissue



(a) Skeletal muscle

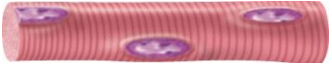
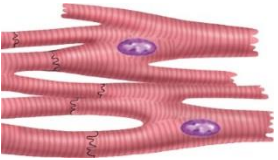


(b) Cardiac muscle



(c) Visceral smooth muscle

Three Types of Muscular Tissue

	Location	Function	Appearance	Control
Skeletal 	skeleton	movement, heat, posture	striated , multi-nucleated (eccentric), fibers parallel	voluntary
Cardiac 	heart	pump blood continuously	striated , one central nucleus	involuntary
Visceral (smooth muscle) 	G.I. tract, uterus, eye, blood vessels	Peristalsis, blood pressure, pupil size, erects hairs	no striations , one central nucleus	involuntary

Muscular System

- **Skeletal muscle is the only organ of the muscular system**
- **Skeletal muscle is composed of skeletal muscle tissue and also contains nervous tissue, blood vessels and connective tissue**
- **Half of the body's weight is muscle tissue**
 - **Skeletal muscle = 40% in males, 32% in females**
 - **Cardiac muscle = 10%**

Skeletal muscle fibers (cells) are arranged into bundles called fascicles

Fascicles are bound by connective tissue

Four different connective tissue coverings

Deep fascia

Surrounds entire skeletal muscle and extends beyond its length

Epimysium

Closely surrounds skeletal muscle, binds fascicles together

Perimysium

Surrounds each fascicle

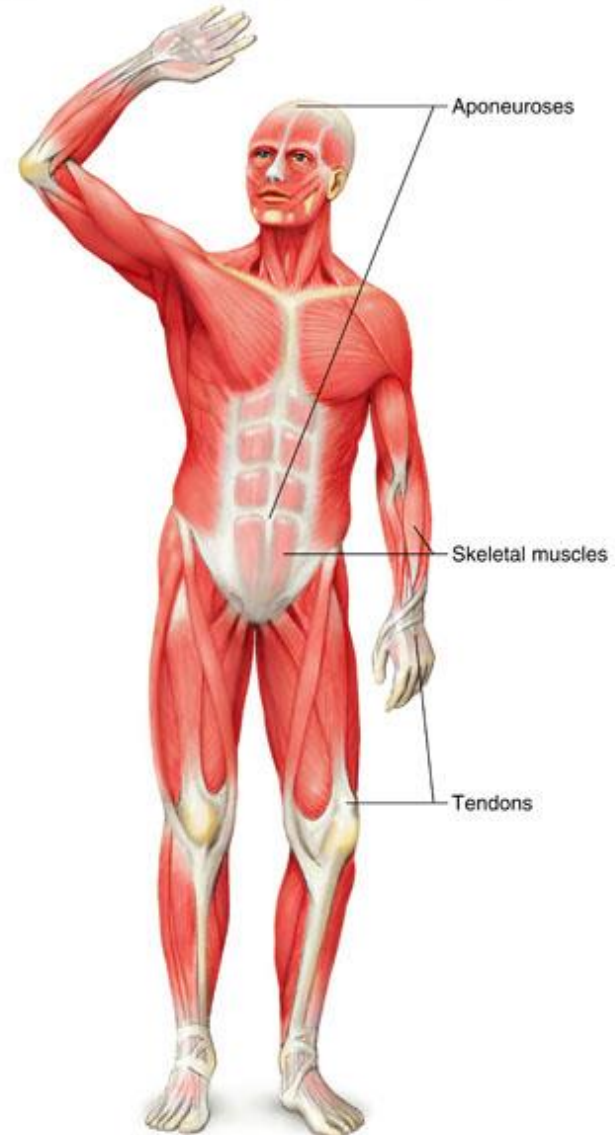
Endomysium

Surrounds each muscle fiber (cell)

Connective Tissue Coverings

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- **Fascia**
 - Surrounds an individual skeletal muscle, separating it from other muscles
 - Fascia may extend beyond the ends of the muscle to become a tendon
 - Fascia may connect muscle to muscle and is called an aponeurosis

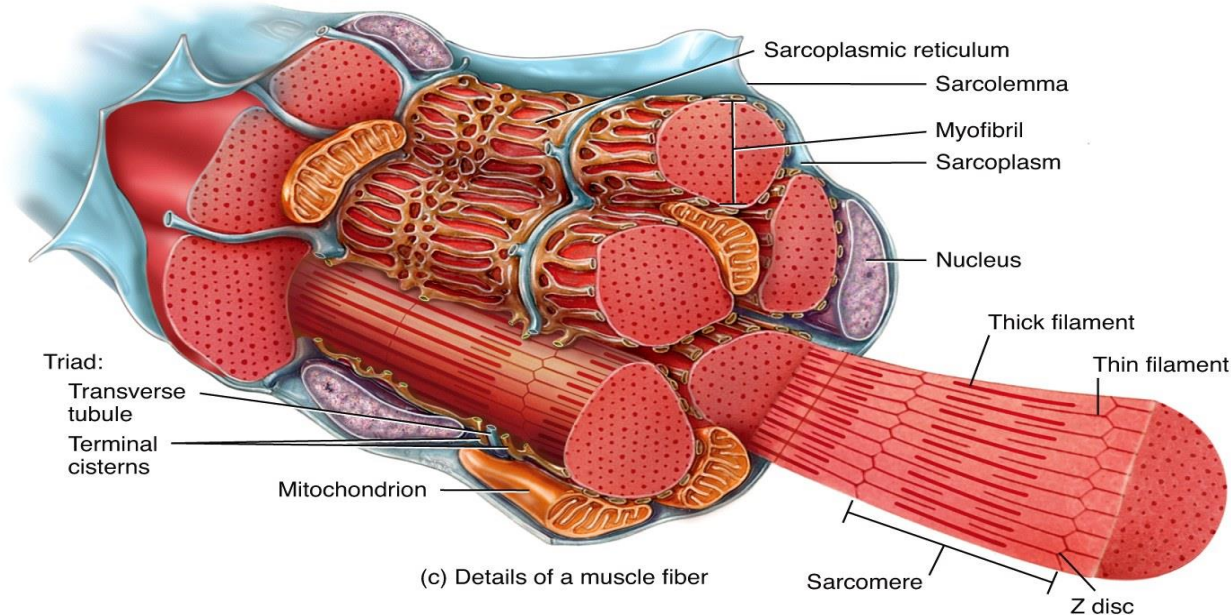


Characteristics of Skeletal Muscle Tissue

- Long, thin contractile fibers that are Striated
 - Striations due to arrangement of thick and thin filaments
- *Under voluntary control*
- Attached to the bones of the skeleton by tendons
- Allow for movement, facial expressions, breathing, swallowing, writing, talking and singing, posture, heat production, joint stability

Skeletal Muscle Arrangement

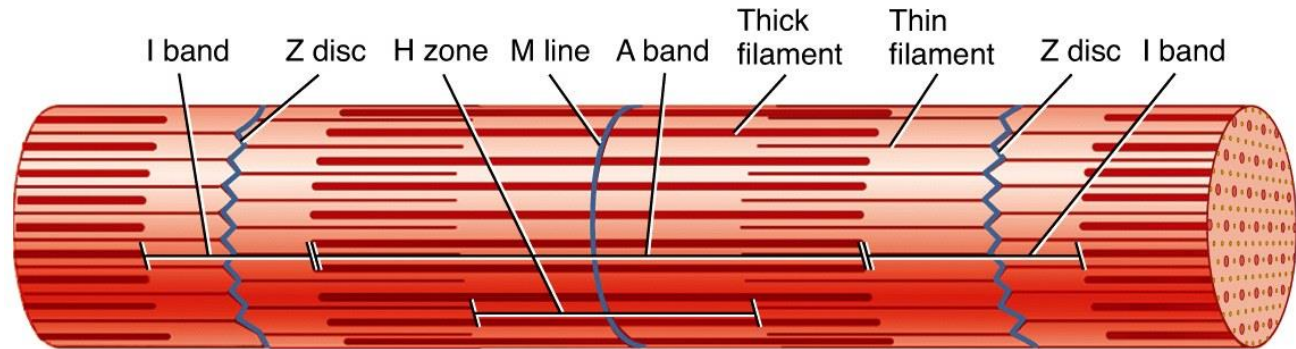
- A single muscle cell is a muscle fiber
 - Fibers are made up of myofibrils
 - Myofibrils are made up of thick and thin filaments
- **Sarcolemma** – muscle cell membrane
- **Sarcoplasm**
- muscle cell cytoplasm



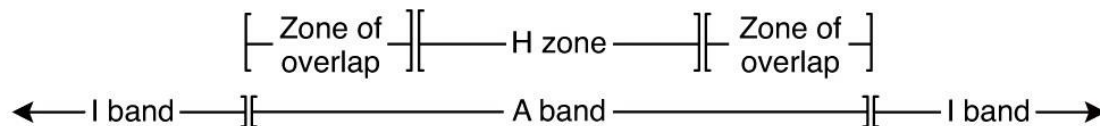
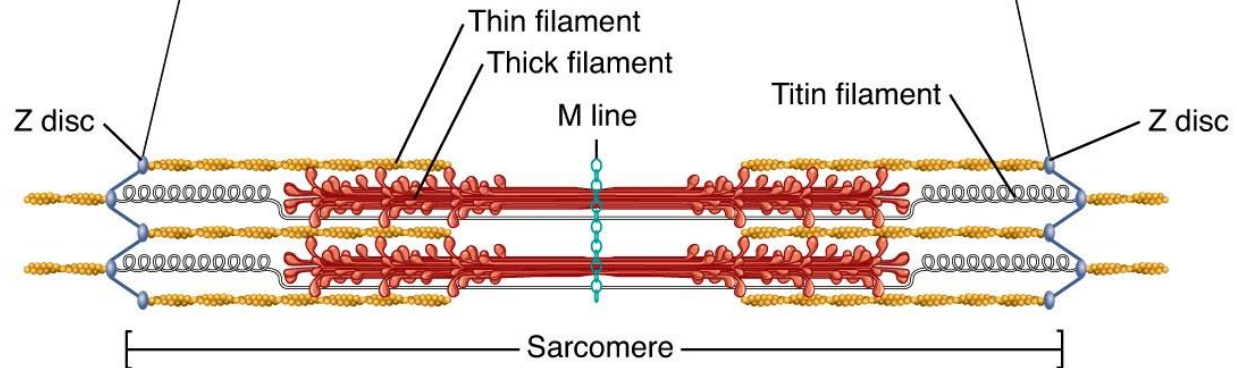
Skeletal Muscle Cells

- Myofibrils are striated
 - Striations due to arrangement of thick and thin filaments
 - Seen as alternating areas of light and dark bands
- The length of each myofibril is divided into repeating units called sarcomeres
 - A sarcomere is the functional unit of skeletal muscle

Sarcomere Arrangement



(a) Myofibril



(b) Details of filaments and Z discs

Sarcomere Structure

- Sarcomere exists from Z-line to Z-line
- **A-Band** is dark middle band
 - Overlapping thick and thin filaments
- **I-Band** – ends of A-Band, thin filaments only
- Z-line is in the middle of the I-Band
- Myosin filaments are held to the Z-line by **titin proteins**

Specialized Organelles of Skeletal Muscle

- **Sarcoplasmic Reticulum (SR)** – a type of ER
 - Surrounds each myofibril, running parallel to it
 - Stores calcium, when stimulated, calcium diffuses into sarcoplasm
- **Transverse Tubules (TT)**
 - Extends into sarcoplasm as invaginations continuous with sarcolemma
 - T tubules run between cisternae of SR
 - Filled with extracellular fluid
 - Cisternae of SR and TT form a triad near where thick and thin filaments overlap

Skeletal Muscle Contraction

Physiology of Skeletal Muscle Contraction: Neurotransmitter Release

- Motor impulse is initiated in the brain
- Travels through the brain and spinal cord to a motor nerve ending
- Motor nerve endings (axons) depolarize
- Calcium enters the axonal endings
- *Calcium causes the release of acetylcholine into the neuromuscular junction (synaptic cleft)*

Physiology of Skeletal Muscle Contraction: Depolarization

- ACh binds receptors on motor end plate
- Depolarizes skeletal muscle fibers
 - Reverses charge of membrane
- Impulse travels through transverse tubules to reach all of the muscle fibers
- *Muscle depolarization causes release of calcium from the sarcoplasmic reticulum into the sarcoplasm*

Physiology of Skeletal Muscle

Contraction: Power Stroke

- Calcium binds troponin (which is attached to tropomyosin)
- *Moves tropomyosin from the myosin binding sites on actin*
- Myosin crossbridges (heads) bind to actin
 - ATP hydrolysis supplies energy
- Actin is pulled inward towards the center of the sarcomere = **POWER STROKE**
- Sarcomeres shorten as muscle contracts

Muscle Relaxation Mechanism

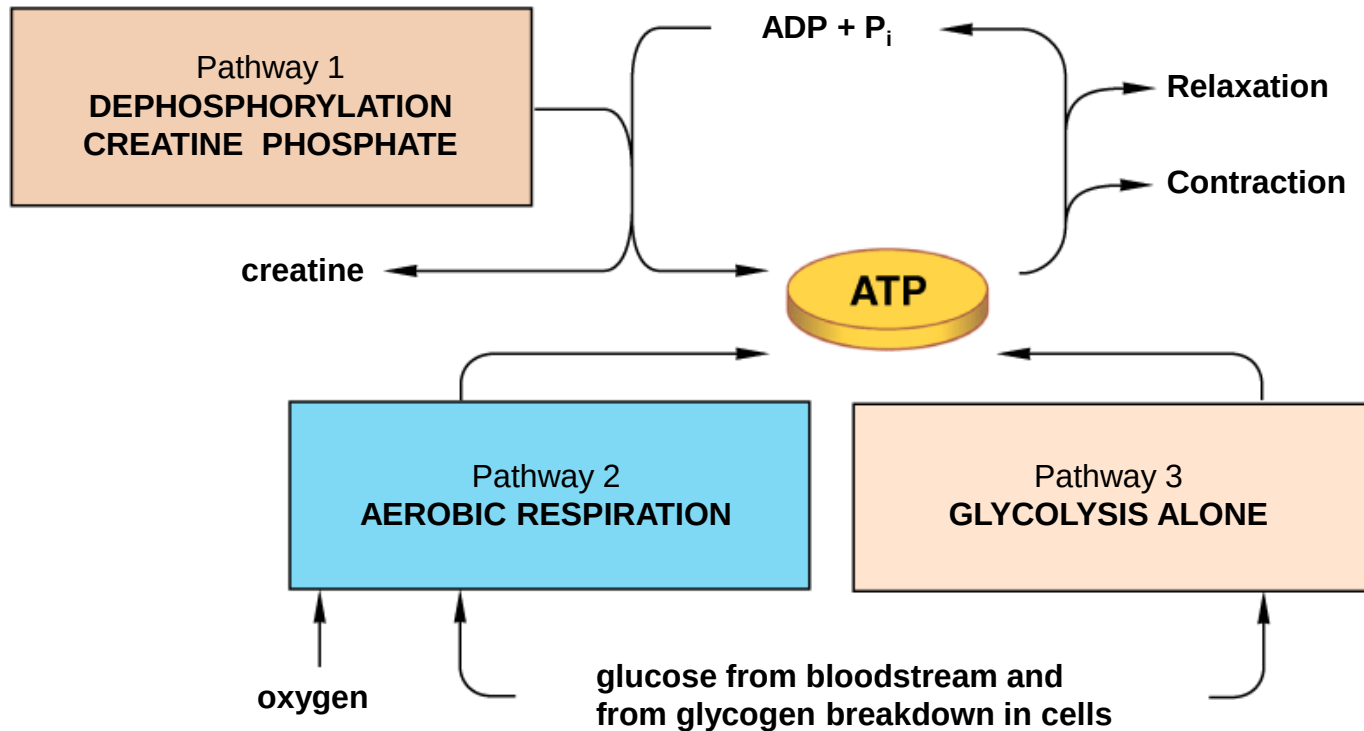
1. Acetylcholinesterase present in the NMJ destroys ACh (preventing continual stimulation)
 2. Calcium ions are transported from the sarcoplasm back into the SR
 3. Linkages between myosin and actin are broken
 - Requires ATP binding
- THEN: The muscle fiber relaxes

Energy for Contraction

- Muscle cells require huge amounts of ATP energy to power contraction
- The cells have only a very small store of ATP
- *Three pathways supply ATP to power muscle contraction*

ATP Supply for Contraction

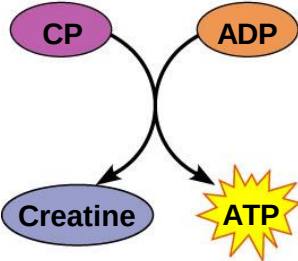
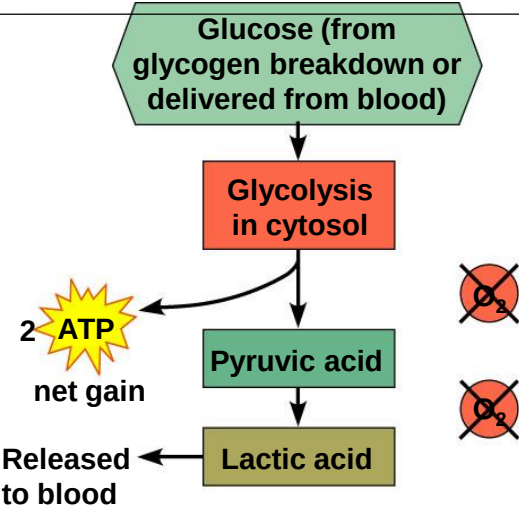
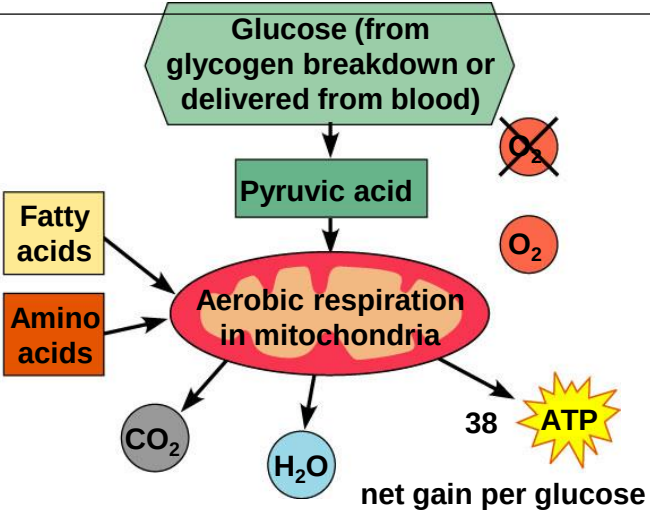
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Energy for Contraction

- ATP initially supplied from cellular respiration
- If ATP is abundant, is converted to creatine phosphate and stored in skeletal muscles
- When ATP is low, creatine phosphate supplies phosphate to ADP making ATP
- CP & ATP stores only good for about a 10 second maximal contraction
- ATP must then come from cellular respiration or glycolysis

Methods of regenerating ATP during muscle activity

		
(a) Direct phosphorylation [coupled reaction of creatine phosphate (CP) and ADP]	(b) Anaerobic mechanism (glycolysis and lactic acid formation)	(c) Aerobic mechanism (aerobic cellular respiration)
Energy source: CP	Energy source: glucose	Energy source: glucose; pyruvic acid; free fatty acids from adipose tissue; amino acids from protein catabolism
Oxygen use: None Products: 1 ATP per CP, creatine Duration of energy provision: 15s	Oxygen use: None Products: 2 ATP per glucose, lactic acid Duration of energy provision: 30–60 s.	Oxygen use: Required Products: 38 ATP per glucose, CO₂, H₂O Duration of energy provision: Hours

Oxygen & Muscle Contraction

- Myoglobin of muscle (similar to hemoglobin) binds to and stores oxygen
- Supplies O₂ needed to make ATP for contraction

Exercise and Skeletal Muscles

- Prolonged, moderate exercise
 - ATP supplied through cellular respiration
 - Once glycogen stores are depleted in muscle, glucose and fatty acid deliveries from blood are used as fuel source

Exercise and Contraction

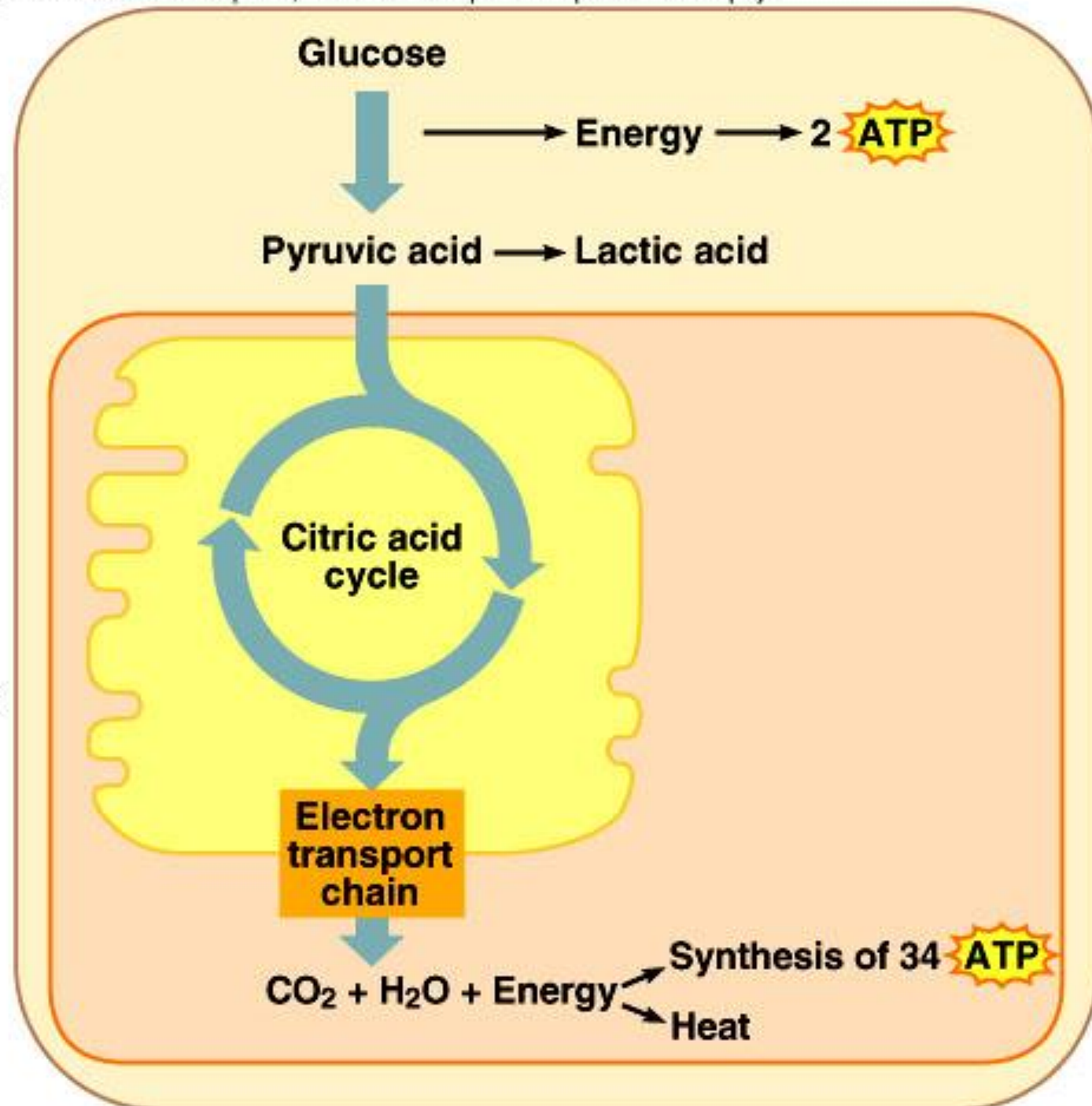
- **In intense strenuous activity, oxygen can be depleted**
- **Contraction of skeletal muscles decreases blood delivery to muscles**
- **Nutrient and O₂ levels in contracting muscles decrease**

Exercise and Skeletal Muscles

- Intense, strenuous exercise
 - Muscles exceed capacity of respiratory and cardiovascular systems to deliver oxygen for contraction
 - ATP supplied anaerobically through glycolysis
 - Pyruvate is converted to lactic acid
 - Lactic acid builds up in muscles
 - Causes muscles to fatigue

1 In the absence of sufficient oxygen, glycolysis leads to lactic acid accumulation.

2 Oxygen carried from lungs by hemoglobin in red blood cells is stored in muscle cells by myoglobin and is available to support aerobic respiration.



Heat Production

- Cellular respiration is only about 40% efficient
- About 60% of the energy found in a glucose is lost as heat during cellular respiration
- Muscle contraction generates heat because muscles use large amounts of nutrients to make ATP, generating large amounts of heat
- Heat is used to maintain body temperature



Cardiovascular System

Cardiovascular System

- A closed system of the heart and blood vessels
 - The heart pumps blood
 - Blood vessels allow blood to circulate to all parts of the body
- The function of the cardiovascular system is to deliver oxygen and nutrients and to remove carbon dioxide and other waste products

The Heart

- Location
 - Thorax between the lungs
 - Pointed apex directed toward left hip
- About the size of your fist
 - Less than 1 lb.

The Heart

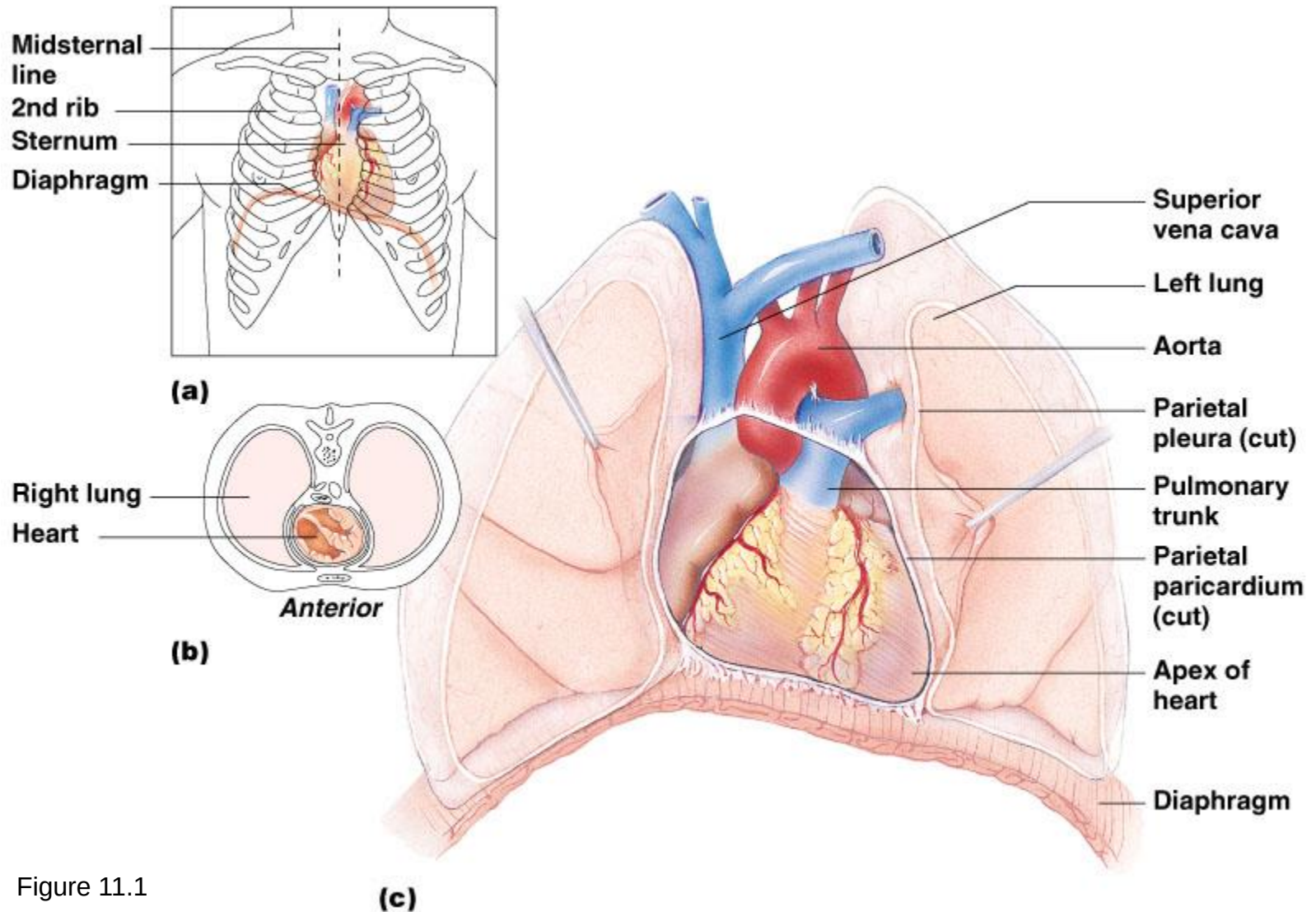


Figure 11.1

The Heart: Coverings

- Pericardium – a double serous membrane
 - Visceral pericardium
 - Next to heart
 - Parietal pericardium
 - Outside layer
- Serous fluid fills the space between the layers of pericardium

The Heart: Heart Wall

- Three layers
 - Epicardium
 - Outside layer
 - This layer is the parietal pericardium
 - Connective tissue layer
 - Myocardium
 - Middle layer
 - Mostly cardiac muscle
 - Endocardium
 - Inner layer
 - Endothelium

External Heart Anatomy

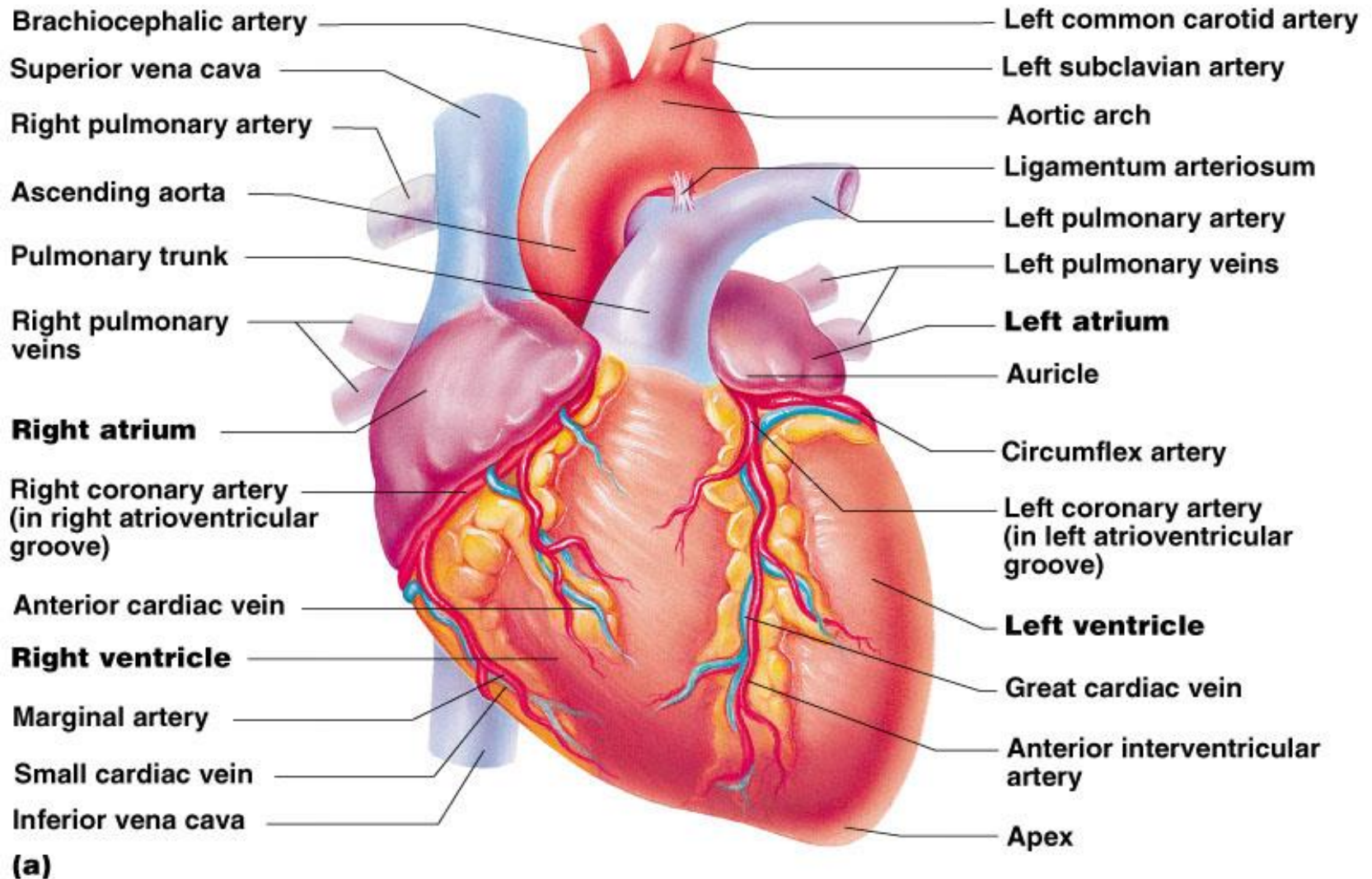


Figure 11.2a

The Heart: Chambers

- Right and left side act as separate pumps
- Four chambers
 - Atria
 - Receiving chambers
 - Right atrium
 - Left atrium
 - Ventricles
 - Discharging chambers
 - Right ventricle
 - Left ventricle

Blood Circulation

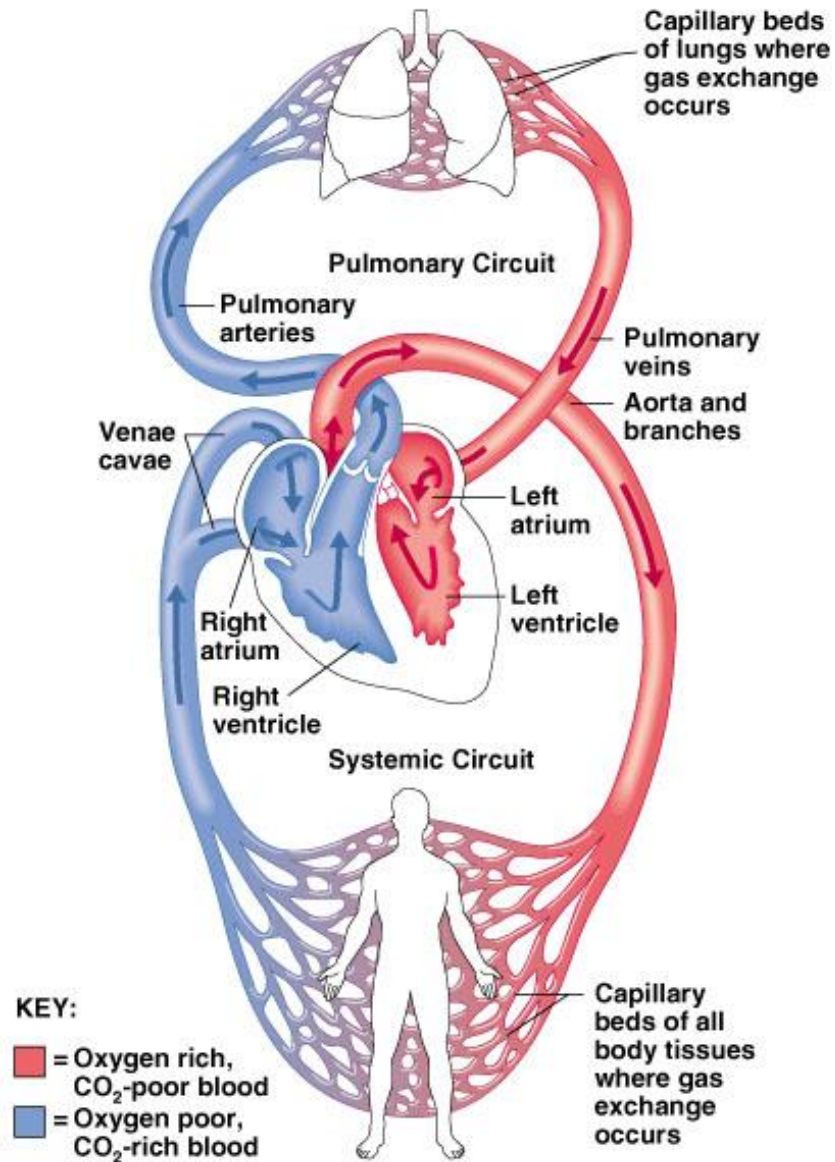


Figure 11.3

The Heart: Valves

- Allow blood to flow in only one direction
- Four valves
 - Atrioventricular valves – between atria and ventricles
 - Bicuspid valve (left)
 - Tricuspid valve (right)
 - Semilunar valves between ventricle and artery
 - Pulmonary semilunar valve
 - Aortic semilunar valve

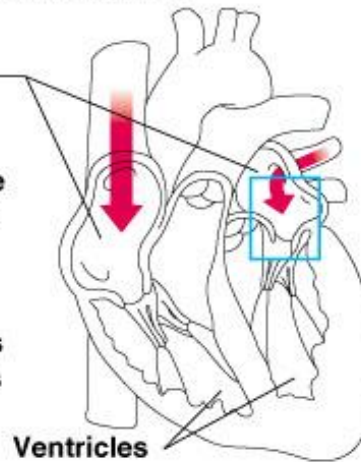
The Heart: Valves

- Valves open as blood is pumped through
- Held in place by chordae tendineae (“heart strings”)
- Close to prevent backflow

Operation of Heart Valves

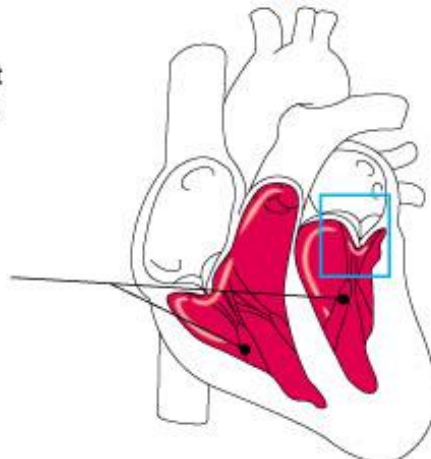
Operation of the AV valves

- ① Blood returning to the heart fills atria, putting pressure against atrioventricular valves; the atrioventricular valves are forced open
- ② As the ventricles fill, atrioventricular valve flaps hang limply into ventricles
- ③ Atria contract, forcing additional blood into ventricles



AV valves open

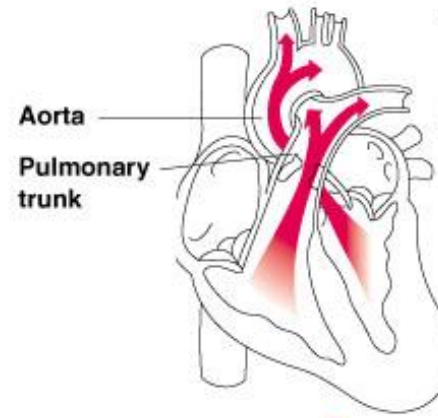
- ① Ventricles contract, forcing blood against atrioventricular valve cusps
- ② Atrioventricular valves close
- ③ Chordae tendineae tighten, preventing valve flaps from everting into atria



AV valves closed

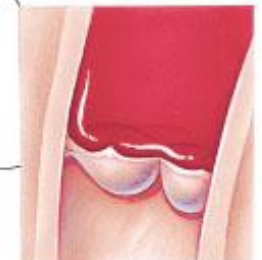
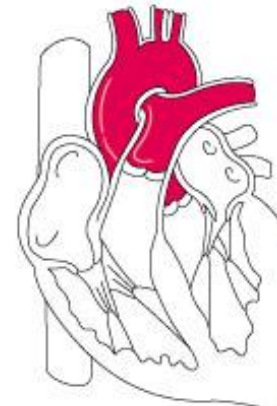
Operation of the semilunar valves

As ventricles contract and intraventricular pressure rises, blood is pushed up against semilunar valves, forcing them open



Semilunar valve open

As ventricles relax, and intraventricular pressure falls, blood flows back from arteries, filling the cusps of semilunar valves and forcing them to close



Semilunar valve closed

Figure 11.4

(a)

(b)

Valve Pathology

- Incompetent valve = backflow and repump
- Stenosis = stiff= heart workload increased
- May be replaced
- Lup Dub Heart Sound

The Heart: Associated Great Vessels

- Aorta
 - Leaves left ventricle
- Pulmonary arteries
 - Leave right ventricle
- Vena cava
 - Enters right atrium
- Pulmonary veins (four)
 - Enter left atrium

Coronary Circulation

- Blood in the heart chambers does not nourish the myocardium
- The heart has its own nourishing circulatory system
 - Coronary arteries
 - Cardiac veins
 - Blood empties into the right atrium via the coronary sinus